

Submanifolds

1.1 Embedded Submanifolds

1.1.1 Slice Charts for Embedded Submanifolds

Definition 1.1.1

If U is an open subset of $\mathbb{R}^n$ and $k \in \{0, \dots, n\}$, a **k -dimensional slice of U** (or simply a **k -slice**) is any subset of the form

$$S = \{(x^1, \dots, x^k, x^{k+1}, \dots, x^n) \in U : x^{k+1} = c^{k+1}, \dots, x^n = c^n\}$$

for some constants $c^{k+1}, \dots, c^n$.

Definition 1.1.2 ► local k -slice condition

Let M be a smooth n -manifold.

- Let (U, φ) be a smooth chart on M . If S is a subset of U such that $\varphi(S)$ is a k -slice of $\varphi(U)$, then we say that S is a **k -slice of U** .
- Given a subset $S \subseteq M$ and a nonnegative integer k , we say that S satisfies the **local k -slice condition** if each point of S is contained in the domain of a smooth chart (U, φ) for M such that $S \cap U$ is a single k -slice in U .
- Any such chart is called a **slice chart for S in M** , and the corresponding coordinates $(x^1, \dots, x^n)$ are called **slice coordinates**.

1.1.2 Level Sets

Definition 1.1.3

If $\Phi : M \rightarrow N$ is a smooth map.

- A point $p \in M$ is said to be a **regular point of Φ** if $d\Phi_p : T_p M \rightarrow T_{\Phi(p)} N$ is surjective; it is a **critical point of Φ** otherwise.
- A point $c \in N$ is said to be a **regular value of Φ** if every point of the level set $\Phi^{-1}(c)$ is a regular point, and a **critical value** otherwise.

Theorem 1.1.4 ▶ Constant-Rank Level Set Theorem

Let M and N be smooth manifolds, and let $\Phi : M \rightarrow N$ be a smooth map with constant rank r . Each level set of Φ is a properly embedded submanifold of codimension r in M .

Proof. Write $m = \dim M$, $n = \dim N$, and $k = m - r$. Let $c \in N$ be arbitrary, let S denote the level set $\Phi^{-1}(c) \subseteq M$. From the rank theorem, for each $p \in S$, there are smooth charts (U, φ) centered at p and (V, ψ) centered at $c = \Phi(p)$ in which Φ has a coordinate representation of the form

$$\psi \circ \Phi \circ \varphi^{-1}(x^1, \dots, x^m) = (x^1, \dots, x^r, 0, \dots, 0)$$

and therefore $S \cap U$, then

$$\begin{aligned} q \in S &\iff \Phi(q) = c \iff (x^1(q), \dots, x^r(q), 0, \dots, 0) = (0, \dots, 0) \\ &\iff x^1(q) = \dots = x^r(q) = 0 \end{aligned}$$

Thus

$$S \cap U = \{(x^1, \dots, x^r, x^{r+1}, \dots, x^m) \in U : x^1 = \dots = x^r = 0\}$$

is the k -slice. So S satisfies the local k -slice condition, so it is an embedded submanifold of dimension k . □